SHRESTH ROHAN

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PROFESSIONAL SUMMARY

Computer Science undergraduate specializing in Data Science with hands-on experience building full-stack AI/ML applications and embedded IoT systems. Skilled in Python, React, Node.js, and applied deep learning, with a filed patent for an IoT-based smart infrastructure system and a strong problem-solving foundation from competitive olympiads.

EDUCATION

Vellore Institute of Technology — Vellore, Tamil Nadu
Bachelor of Technology in Computer Science, Specialization in Data Science | CGPA: 8.76/10 (through Semester 4) | July 2024 – May 2028 (Expected)

St. Michael's High School — Patna, Bihar
Class XII, CBSE | 90.2% | May 2024

Don Bosco Academy — Patna, Bihar
Class X, ICSE | 96.8% | May 2022

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, JavaScript, HTML/CSS

Frameworks & Libraries: React, Node.js, Express, Spring Boot, Flask

AI / ML: PyTorch, ONNX, MediaPipe, Pandas, NumPy, Matplotlib

Databases: MongoDB, PostgreSQL

Developer Tools: Git, Docker, VS Code

PROJECTS

FocusGuard | Python, PyTorch, React, Node.js, MongoDB, ONNX
Feb 2026 – Present

- Built a full-stack AI attention-monitoring system using MobileNetV2 transfer learning to classify user focus state (focused, distracted, drowsy) in real time from webcam input
- Trained and exported a custom ONNX model, serving live inference through a Flask REST API with MediaPipe face detection and temporal smoothing for stable predictions
- Developed a React dashboard with session analytics, a productivity score, and real-time confidence visualization, backed by a Node.js/Express API and MongoDB
- Built a Chrome extension using the Mutation Observer API to dynamically filter distracting DOM elements on sites including YouTube, Instagram, and Twitter

Smart Adaptive Street Lighting System (Patent Filed) | ESP32, Arduino IDE, C++, IoT Sensors
April 2025 – Present

- Engineered an ESP32-based intelligent street lighting system integrating LDR, rain, sound, and ultrasonic sensors with a servo motor for adaptive illumination based on environmental and traffic conditions
- Implemented dynamic brightness control using ambient light level, rainfall detection, and emergency-siren recognition to improve energy efficiency and prioritize emergency vehicle movement
- Designed a servo-assisted ultrasonic scanning mechanism for vehicle detection, distance measurement, and overspeed monitoring with real-time buzzer alerts for enhanced road safety
- Filed an Indian patent application for the multi-sensor smart street lighting system, covering an integrated approach to adaptive lighting, traffic monitoring, and smart-city infrastructure

ACHIEVEMENTS

- Ranked 3rd in State and 239th internationally in the SilverZone Foundation International Informatics Olympiad
- Achieved Zonal Rank 1 and International Rank 85 in the SOF International English Olympiad

LEADERSHIP & COMMUNITY INVOLVEMENT

- Volunteer, Karo Sambhav Movement — supported e-waste collection and recycling awareness initiatives
- VIT Outreach Program — taught middle-school students the fundamentals of computers and coding

INTERESTS

Game Development (OpenGL, Unreal Engine 5) • UI/UX Design (Figma, Canva) • Competitive Programming